

DELTA ENGINE

Autonomous, Agentic & Self-Healing Event Operating System



DELTA ENGINE

PROUDLY PRESENTED BY TEAM DELTA

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Live Production URL: <https://deltaengine.vercel.app/>

GitHub Repository: <https://github.com/aryanpandeyspec-cyber/DELTA-engine>



Chaos & Operational Bottlenecks in Event Infrastructure

■ Legacy Conference Failures & Flaws

- **Cascading Speaker Delay Bottlenecks:** Traffic delays or delayed flights cause speaker arrivals to shift unpredictably. In legacy platforms, a 30-minute keynote delay freezes the entire track, causing domino cancellations across parallel halls.
- **Dangerous Room Capacity Surge:** High-interest keynote topics draw 200+ attendees into 60-seat rooms, creating severe crowd congestion, fire safety risks, and broken audio/visual coverage.
- **Manual Coordination Chaos:** Event organizers waste critical hours frantically calling speakers, swapping HDMI cables, and reprinting paper schedules that become obsolete within minutes.
- **Stale Static Telemetry:** Attendees, speakers, and venue staff rely on static schedule boards with zero visibility into live schedule shifts or room reallocations.

■ The Imperative for an Event OS

- **Autonomous Decision Engine:** Systems must resolve timing & hall constraints instantly without human bottlenecks or manual delay lag.
- **Multi-Agent LLM Reasoning:** Heterogeneous AI models must negotiate schedule fixes peer-to-peer in real time to produce human-readable resolution rationale.
- **Real-Time Telemetry:** Low-latency bi-directional WebSocket synchronization across thousands of live attendee & staff displays.
- **Automated Emergency Dispatch:** Instant DKIM email & WhatsApp notifications sent directly to stage leads, mic runners, and door coordinators.

30+ Mins

Average Delay Cascade in Legacy Events

200+ Attendees

Swarming 60-Cap Halls Without Solvers

100% Zero

Real-Time Telemetry in Static Board Systems



DELTA ENGINE: The Self-Healing Event Operating System

■ ■ Self-Healing Solver Engine

- **Iterative Pass Solver:** Evaluates up to 10 constraint passes automatically under peak concurrent load.
- **Speaker Availability Formula:** Auto-adjusts start times based on live minutes delay telemetry.
- **Capacity Surge Room Swap:** Swaps venue halls dynamically when attendee interest breaches room capacity.
- **Deterministic Stabilization:** Stabilizes matrix parameters if constraints reach boundary limits.

■ 4-Agent LLM Swarm (Groq)

- **Specialized Groq AI Models:** Liaison (Llama-3.1-8B), Scheduler (Llama-3.3-70B), Logistics (70B), Marketing (8B).
- **Parallel Reasoning Pipeline:** Scheduler & Logistics execute concurrently via Promise.all.
- **Circuit Breaker Mode:** Super Admin token limiter toggle for API cost control & safety.
- **Human-Readable Rationale:** Generates multi-role dialogue explaining resolution steps.

■ Dynamic Graph & Dispatch

- **In-Memory Relational Graph:** Models Speakers, Topics, Halls, Slots & Directional Edges.
- **Deterministic Date Seeding:** Hash algorithms seed unique non-overlapping schedules per calendar date.
- **Multi-Channel Dispatch:** Instant WhatsApp AI logs & DKIM-signed Supabase email notifications.
- **Low-Latency WS Broadcast:** Pushes full graph updates in real time to all connected clients.

< 150 ms

Autonomous Conflict Healing Latency

4 Parallel LLMs

Groq Heterogeneous Agent Swarm

100% Automated

Multi-Channel Emergency Dispatch



End-to-End Autonomous Data & Decision Flow Pipeline

■ ■ Multi-Layer Execution Pipeline Breakdown

- **Step 1 — Sensor Telemetry Ingestion Layer:** Disruption alerts (e.g. Speaker 1 delayed by 30 mins or Topic 2 interest surge to 210 in 120-cap hall) ingested via REST API or UI state change.
- **Step 2 — Algorithmic Self-Healing Layer:** backend/components/selfHealing.js audits availability windows and venue capacity bounds, computing optimal slot shifts and room swaps.
- **Step 3 — Multi-Agent AI Swarm Layer:** backend/components/agentSwarm.js executes Groq LLMs in a hybrid parallel-sequential pipeline to generate human-readable resolution dialogues.
- **Step 4 — Graph Edge Resynchronization:** backend/components/graphDb.js rebinds directional graph edges (SCHEDULED_IN, SCHEDULED_AT) in memory.
- **Step 5 — WebSocket Telemetry Broadcast:** backend/server.js pushes JSON state payloads (INIT_STATE, SELF_HEAL_UPDATE) to all connected client dashboards in real time.
- **Step 6 — Multi-Channel Emergency Dispatch:** Auto-dispatches DKIM-signed Supabase HTML emails and WhatsApp AI liaison logs to affected coordinators and speakers.

Layer 1-2

Telemetry & Algorithmic Solver

Layer 3-4

AI Swarm & Graph Edge Resync

Layer 5-6

WebSocket Sync & Email/WA Dispatch



Deep-Dive: Autonomous Constraint Solvers

■ ■ Speaker Delay Solver Algorithm

- **Availability Start Formula:** Availability Start Hour = $9.0 + (\text{Delay Minutes} / 60)$.
- **Conflict Condition:** Raised if Scheduled Start Hour < Availability Start Hour.
- **Slot Shift Heuristic:** Searches subsequent slots for open availability or shifts lower-interest sessions to later hours.

```
resolveSpeakerDelay(topicId, slotId, hallId, startHour,  
logs, notifications, db)
```

■ ■ Room Capacity Overflow Solver

- **Capacity Matrix Audit:** Evaluates Topic Interest Score against Hall Capacities (Turing Hall = 250, Lovelace Suite = 120, Hopper Room = 60).
- **Smart Venue Swap:** Identifies parallel sessions with lower attendance in larger halls and executes a clean venue swap.
- **Bounded Solver Safeguard:** Bounded to 10 iterations max to guarantee zero infinite loops under heavy concurrent stress.

```
resolveCapacityOverflow(topicId, slotId, hallId, logs,  
notifications, db)
```

9.0 + (Delay/60)

Exact Availability Start Formula

10 Passes Max

Bounded Iteration Safeguard Limit

Turing: 250 Cap

Dynamic Seating Allocation Matrix



Deep-Dive: Heterogeneous 4-Agent LLM Swarm (Groq AI)

■ ■ Liaison Agent

- **Model:** llama-3.1-8b-instant
- **Role:** Raw disruption sensor telemetry ingestion.
- **Output:** Ultra-fast 1-sentence status report.

■ ■ Scheduler Agent

- **Model:** llama-3.3-70b-versatile
- **Role:** Slot timing & schedule constraints.
- **Execution:** Parallel via Promise.all.

■ ■ Logistics Agent

- **Model:** llama-3.3-70b-versatile
- **Role:** Venue capacity & AV gear audit.
- **Execution:** Parallel via Promise.all.

■ Marketing Agent

- **Model:** llama-3.1-8b-instant
- **Role:** Formulates public push alerts.
- **Output:** Final push alert & iCal sync.

■ ■ Super Admin Circuit Breaker / Token Limiter Mode

- **Token Cost & Fallback Guard:** Toggle in agentSwarm.js blocks external API token expenditure during traffic surges, automatically dropping back to a deterministic rule-based optimization solver without interrupting live operations.



Deep-Dive: Dynamic Relational Graph Database

■ In-Memory Relational Graph Schema

- **Speakers Nodes:** ID, name, role, bio, avatar, live minute delay tracker.
- **Topics Nodes:** Title, speaker ID, tags, interest score, slide deck upload status.
- **Halls & Slots Nodes:** Room capacities and start/end time windows.
- **Directional Edges:** SPEAKER_OF, SCHEDULED_IN, SCHEDULED_AT.

■ Multi-Date Deterministic Seeding

- **ASCII Character Hash:** Date string (e.g., 2026-08-07) converted to numeric seed value.
- **Multi-Day Event Support:** Generates unique non-overlapping schedule matrices per day automatically.
- **Live Edge Resync:** Switching calendar dates dynamically updates active graph edges.
- **Volunteer Assignment:** Maps stage leads, mic runners, and door scanners to active topics.

4 Node Types

Speakers, Topics, Halls, Time Slots

3 Edge Types

SPEAKER_OF, SCHEDULED_IN,
SCHEDULED_AT

ASCII Seed

Deterministic Multi-Date Hash Engine



Automated Communication & Emergency Dispatch

■ Supabase Anti-Spam Email Engine

- **Automated HTML Templates:** Composes responsive HTML emails containing session reallocation notices, dispatch hashes, timestamping, and original vs reallocated venue parameters.
- **High Deliverability:** Styled for DKIM/SPF compliance to prevent spam filtering.
- **Supabase DB Persistence:** Records dispatches in event_email_dispatches table.
- **Resilient Fallback Mode:** Local cache mode ensures zero email log loss during network drops.

■ WhatsApp AI & Volunteer Registry

- **WhatsApp AI Dispatcher:** Dispatches structured WhatsApp logs directly to stage leads and room coordinators.
- **WebSocket Broadcast Hub:** Dispatches render instantly in client dashboard feeds.
- **Volunteer Duty Registry:** Duty tracking for Rohan Sharma (Stage Lead), Priya Patel (Door Scanner), Aarav Mehta (Mic Runner), Ananya Sen (AV Monitor).

DKIM / SPF

High-Deliverability Anti-Spam Email

WhatsApp AI

Automated On-Ground Dispatcher

Local Cache

Resilient Fallback Mode Guard



Super Admin Command Center & System Reliability

■ Super Admin Command Center

- **LLM Token Circuit Breaker:** Toggle to pause AI spending during high traffic surges.
- **500-Request Stress Test Suite:** Built-in load testing engine benchmarked in stressTester.js.
- **Graph Node Canvas Inspector:** Interactive canvas inspector for real-time physics nodes.
- **Live Telemetry Monitor:** Real-time WebSocket connection state and memory monitoring.

■ Security & Process Crash Guards

- **Process Uncaught Exception Guards:** Node listeners prevent process termination on unhandled errors.
- **Express Rate Limiter:** Protects API endpoints against Denial of Service (DoS) floods.
- **Strict File Buffer Validation:** Multer memory storage enforces strict 20MB limit & format checks.
- **HTML Entity Sanitization:** All user inputs sanitized to prevent XSS vector attacks.

500 Requests

High-Concurrency Stress Test Suite

Crash Guard

Process Uncaught Exception Listener

20 MB Max

Strict File Buffer Validation Guard



Live Demonstration, Scalability & Roadmap

■ Real-World Impact & Production Deployment

- **Live Production URL:** Deployed on Vercel at <https://deltaengine.vercel.app/>
- **Real-World Impact:** Eliminates event scheduling delays, prevents room overcrowding, and automates coordinator workflows end-to-end.
- **Performance Metrics:** Solves multi-variable schedule conflicts in <150ms; handles 500 concurrent telemetry requests cleanly.
- **Scalability & Roadmap:** Integration with IoT room occupation sensors, automated SMS gateways, and direct Google Calendar / Apple Wallet iCal sync.
- **Thank You from Team DELTA:** Aryan Pandey, Shaik Shahiduddin, Suryansh Singh (2nd Year Students).

Vercel Live

<https://deltaengine.vercel.app/>

Team DELTA

Aryan, Shahiduddin, Suryansh

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HackIndia Finalist Project